HIDDEN MARKOV MODELS FOR HUMAN ACTIVITY RECOGNITION FROM MOTION SENSOR DATA

A Report on Accelerometer-Based Activity Classification Using HMM

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1. Background and Motivation

Being able to recognize human activity is an important area for some areas of study, including health care monitoring, fitness, and wearable technology. Most modern smartphones have motion sensors like an accelerometer or a gyroscope built into them that can accurately collect motion data for an extended length of time; however, you cannot directly see the actual activity you are doing (e.g., whether you are walking or standing). Instead, you have to infer your activity based on your overall performance and from previous recorded measurements taken from these sensors. A Hidden Markov Model (HMM) would be an effective solution for investigating these types of issues, as HMMs are specifically designed to represent multiple sequences, where the actual state is hidden but can still affect the resulting output values. This paper demonstrates the application of Hidden Markov Models to infer human activity from motion sensor outputs as obtained from the use of acceleration measurement systems. The purpose of the project was to develop a functional solution that allows a person to be tracked through the smartphone using the motion sensors for the purpose of monitoring their health (such as weight loss or gain) and for use by elderly individuals, as well as in the monitoring of ward behaviour through an elderly care system.

2. Data Collection and Preprocessing Steps

2.1 Data Collection and Preprocessing

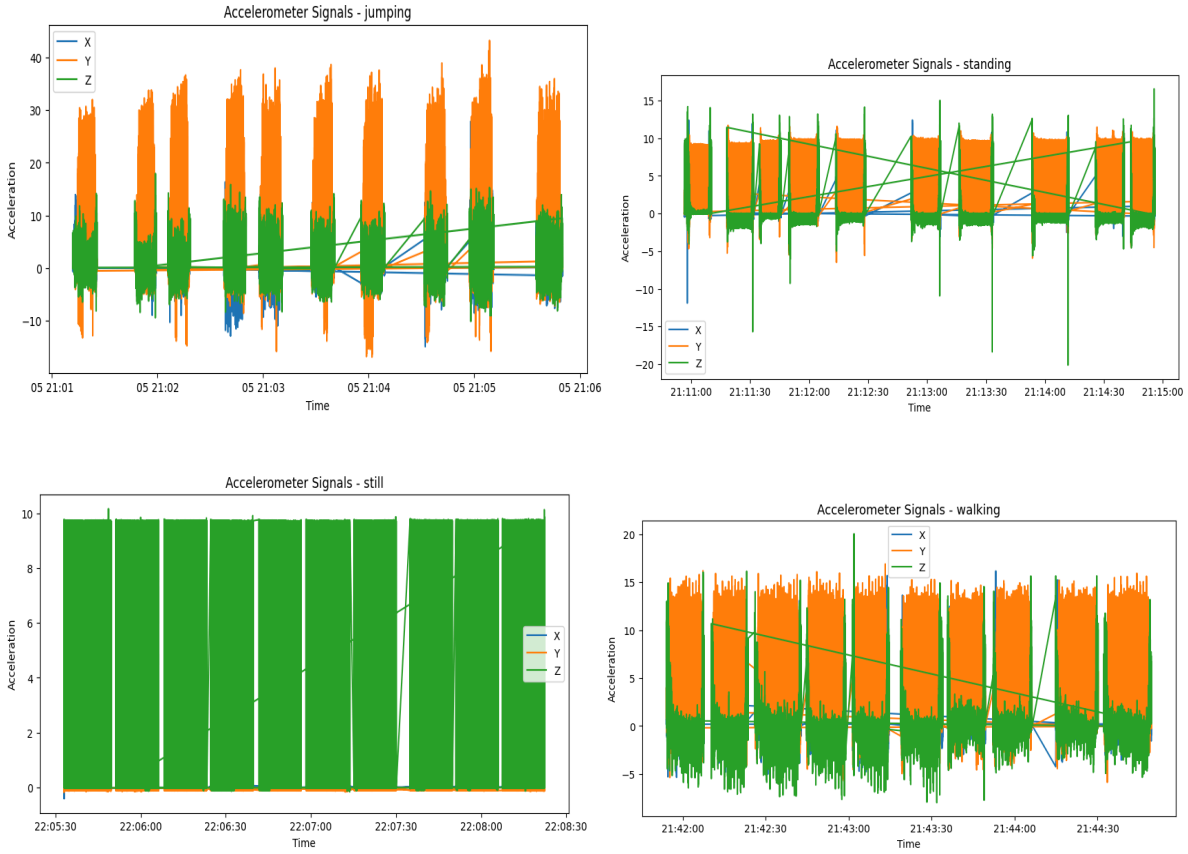
A smartphone motion logging application recorded motion data from three separate points in its path. Four different types of activities were recorded, where either the phone was in a fixed position at waist level (standing), or at normal speed (walking), repeatedly (jumping), and then on a flat surface (still). Each activity was captured for a few seconds to create about fifty recordings total, with time-stamped measurements of each type of activity saved as CSV files uploaded to Kaggle to provide a reproducible programmatic access point in the analysis notebook.

Several preprocessing steps were completed to prepare the data for use within the model. Initially, all duplicate timestamps were removed, then all data was sorted in chronological order prior to removing any missing values from the data. Each timestamp was converted to a datetime format before the magnitude of acceleration was calculated by the following formula:

$$acc_mag = \sqrt{x^2 + y^2 + z^2}$$

This feature helps represent overall movement intensity regardless of device orientation.

The sampling rate was estimated by calculating the average time difference between consecutive timestamps. This step ensured that data from different recordings were consistent and suitable for time-series analysis.



These visualizations help confirm that different activities produce different motion patterns.

2.2 Feature Extraction

Since the data was collected using different phones, the sampling rates were slightly different (48–50 Hz). To harmonize the datasets, the timestamps were used to estimate the effective sampling frequency, and the data were resampled to a consistent rate of **50 Hz** before feature extraction. This ensured that sliding windows represented the same time duration across all recordings. Using overlaps preserves temporal continuity and also allows for more training samples to be available for the model later. From each window, a set of time-domain and frequency-domain features was extracted, which characterise the distinct motion patterns that are associated with different activities. The time-domain features that were obtained from each time series include mean, variance, standard deviations, signal magnitude area (SMA), correlations between x and y, x and z, and y and z axes; maximum, minimum, range; interquartile range (IQR); and root mean square (RMS). These time domain measures represent the statistical distribution, variability, and intensity of the motion.

Frequency-domain features were obtained from the Fast Fourier Transformation (FFT) of the motion signal. These frequency-domain features included the “dominant frequency”, which

identifies the most dominant periodic component of the motion signal, and spectral energy (which quantifies the total power of the signal as measured in the frequency domain). These frequency-domain features will enable us to discriminate based on activity type. Before training the Hidden Markov Model, all features were standardized using StandardScaler. This step ensures that all features have a comparable scale, preventing features with larger numerical ranges from dominating the model and improving the stability of the learning process.

3. HMM Setup and Implementation

3.1 Implementation

The Hidden Markov Model was created to model the patterns exhibited by people while engaging in different types of activities using features derived from accelerometer and gyroscope sensor data. The features extracted from the activity (or motion) data served as inputs (observations) to the HMM, while the different types of activities provided the basis for the hidden states that characterized the underlying dynamics associated with these activities. The activities studied were: standing, walking, jumping, and still. The implementation of the HMM model was done in Python using the hmmlearn package. Before training, the HMM was trained in a multi-class way, and after making a separate model for each type of activity with two hidden states to capture variations in activity of the same type like amount of intensity of activity when walking, each activity-based model then used a diagonal covariance matrix to both decrease the number of parameters learned during training and improve the stability of the models during training.

The Baum-Welch algorithm was used to estimate the HMM parameters. Training stopped when the change in log-likelihood between iterations became smaller than a predefined threshold or when the maximum number of iterations (200) was reached. The trained models were evaluated on unseen data. A sequence was generated for each feature vector and scored against all HMMs. The activity was selected based on which HMM provided the maximum likelihood score. This type of evaluation kept the time relation in the sensor data since the evaluation was made by each movement in sequence against the HMMs. Furthermore, Viterbi decoding was employed to find the best-hidden state path for each model and provide insight into the internal patterns of motion modeled by the HMM.

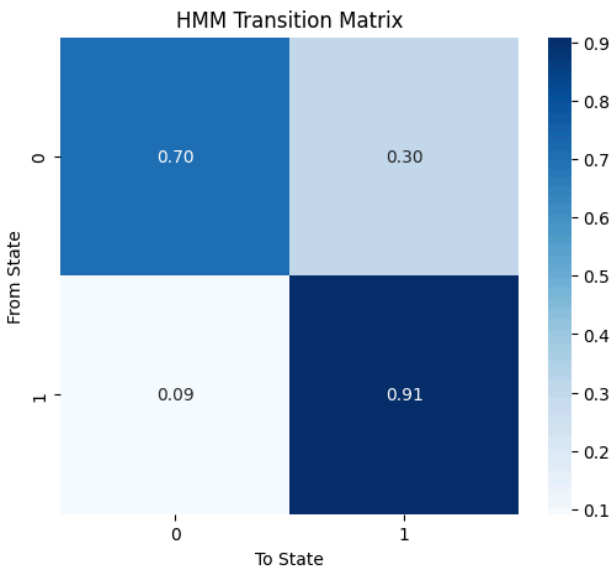
4. Model Evaluation

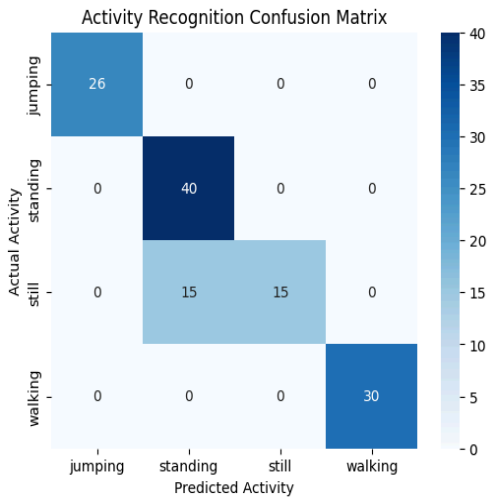
The Hidden Markov Model classifier produced an overall accuracy of 88.10% against previously unseen test data, indicating that it did well in classifying 4 types of movement. The models performed perfectly when classifying both dynamic movements (jumping and walking) and had a precision, recall, and F1-score of 1.0 for those two types of activities. The model's performance declined when it was trying to classify one of the two static postural activities (standing and stillness) since all instances of standing were classified correctly, but

only half of the instances of stillness were classified correctly, and therefore the sensitivity on that type of activity was only 0.50, thus the precision of standing activity was lowered to 0.73 as well. The implication of these results indicates that the model is biased towards identifying standing as an activity whenever it exhibits uncertainty, likely due to the model combining similar features for both of these postural states.

Activity	Support	Sensitivity (Recall)	Precision	Specificity	F1-Score
Jumping	26	1.00	1.00	1.00	1.00
Standing	40	1.00	0.73	0.83	0.84
Still	30	0.50	1.00	1.00	0.67
Walking	30	1.00	1.00	1.00	1.00
Overall					0.88

The confusion between standing and still suggests that the current feature set does not fully capture the subtle differences between static postures. Future improvements could involve incorporating additional features such as accelerometer variance or gyroscope readings, collecting more diverse data for static classes, or tuning the HMM structure, including the number of hidden states per activity. Despite this limitation, the model's high overall F1-score of 0.88 and perfect recognition of dynamic activities confirm its effectiveness as a baseline for human activity recognition, providing a solid foundation for further refinement.





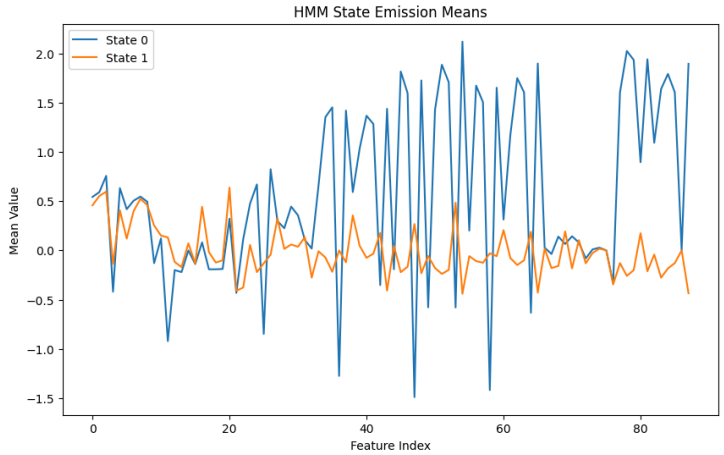


Fig: confusion matrix and Emission probabilities

5. Discussion and Conclusion

5.1 Activity Distinguishability

Similar patterns of motion within an activity indicate the difficulties encountered in distinguishing between motion patterns for the two activities that were frequently misclassified. The errors of classification were apparent in the confusion matrix and also in the similarities in temporal transition of activities and how the hidden Markov model (HMM) was able to capture persistence of activity sequences, but how it had more difficulty in determining where an activity boundary occurred, particularly when both activities had similar frequency and amplitude characteristics. The feature extraction process, in terms of time- and frequency-domain features, provided a substantial amount of useful information for discriminating between activities; however, there were still some activities that could not be readily discriminated using the current methods. The sliding window method and standardized features permitted a consistent input into the HMM.

5.4 Conclusion

This demonstrated that Hidden Markov Models are an effective approach for recognizing human activities from smartphone motion sensor data. The model achieved an overall accuracy of 88.10% and perfect classification of dynamic activities, highlighting the strength of HMMs in capturing temporal patterns and sequential dependencies in motion data. The primary challenge was distinguishing between static postures due to the similarity in their feature representations, resulting in lower sensitivity for stillness and reduced precision for standing.

These results underscore the importance of feature engineering, data diversity, and careful model design when dealing with subtle activity distinctions. Future improvements could include incorporating additional sensor features, expanding the dataset for static activities,

and refining the HMM structure to better capture nuanced motion differences. Despite these limitations, the HMM provides a strong baseline for human activity recognition, offering a foundation for further research and practical applications in health monitoring, elderly care, and fitness tracking.

Appendix

A. Repository Link

Github Link

B. Team Task Sheet

Task Tracker

C. Notebook

Notebook